



Name: Cohort/Batch: Date: Score:

SHARDING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Sharding and what problem in Distributed Systems does it solve?

Q6 How does Sharding handle reliability and high availability?

Q2 What are the core components or concepts in Sharding?

Q7 What observability does Sharding provide out of the box?

Q3 How does Sharding compare to its main alternatives?

Q8 What are common performance bottlenecks in Sharding?

Q4 How do you get started with Sharding for a new project?

Q9 What security best practices apply when running Sharding?

Q5 What configuration options most affect Sharding's behavior in production?

Q10 What mistakes do teams commonly make when adopting Sharding?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Sharding is a tool or platform that

Mitigation

Sharding is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Sharding provides HA through leader election, replication, or stateless horizontal scaling.

Reliability

Sharding provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Sharding has key building blocks that work

Primitives

Sharding has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Sharding exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces.

Observability

Sharding exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Sharding trades off simplicity against feature richness

Hardening

Sharding trades off simplicity against feature richness differently than its alternatives. Choose Sharding when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches.

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Sharding via its package manager or binary release.

Gotchas

Install Sharding via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Sharding with the principle of least privilege

Assessment

Run Sharding with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels.

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Configuration

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.